Name: Date: Period:

Motion #4

$$v = \frac{d}{t} \qquad a = \frac{\Delta v}{t} \qquad \Delta v = v_f - v_i$$

Please use the proper problem-solving method. (1) Draw a picture; (2) knowns & unknowns; (3) pick an equation; (4) plug & chug; (5) answer with units.

1. What is the acceleration of a unicycle that goes from 3 m/s to 5 m/s in 7 s?

2. A train is traveling at 32 m/s and begins slowing down at $-4\,\mathrm{m/s^2}$. What will its velocity be after 5 s?

3. A hiker walks at a constant velocity of 1.5 m/s. How long will it take the hiker to travel **1.6 km**?

$$v = \frac{d}{t} \qquad a = \frac{\Delta v}{t} \qquad \Delta v = v_f - v_i$$

4. What is the velocity of an ATV that travels 13 m in 2 s?

5. A car is traveling at 24 m/s. How far will it travel in **2.5 minutes** if it maintains this velocity?

6. A skateboarder is traveling at 6 m/s and accelerates at a rate of 2.5 m/s^2 for 4.0 s. What is the skateboarder's final velocity?